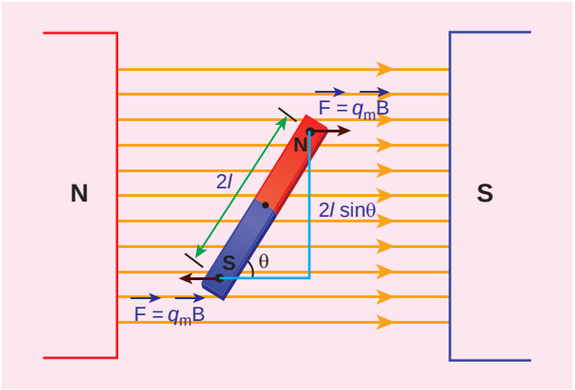


Figure 3.19 Magnetic dipole kept in a uniform magnetic field

This implies, that the net force acting on the dipole is zero, but forms a couple which tends to rotate the bar magnet clockwise (here) in order to align it along $\vec{B}$.

The moment of force or torque experienced by north and south pole about point O is

$$\vec{\tau} = \overrightarrow{ON} \times \vec{F}_N + \overrightarrow{OS} \times \vec{F}_S$$

$$\vec{\tau} = \overrightarrow{ON} \times q_m\vec{B} + \overrightarrow{OS} \times \left(-q_m\vec{B}\right)$$

By using right hand cork screw rule, we conclude that the total torque is pointing into the paper. Since the magnitudes $\left|\overrightarrow{ON}\right| = \left|\overrightarrow{OS}\right| = l$ *and* $\left|q_m\vec{B}\right| = \left|-q_m\vec{B}\right|$, the magnitude of total torque about point O

$$\tau = l \times q_m B\sin\theta + l \times q_m B\sin\theta$$

$$\tau = 2l \times q_m B\sin\theta$$

$$\tau = p_m B\sin\theta \qquad \left(\therefore q_m \times 2l = p_m\right)$$

In vector notation, $\vec{\tau} = \vec{p}_m \times \vec{B}$ (3.25)

EXAMPLE 3.7

Show the time period of oscillation when a bar magnet is kept in a uniform magnetic field is $T = 2\pi\sqrt{\dfrac{1}{p_m B}}$ in second, where I represents moment of inertia of the bar magnet, p_m is the magnetic moment and is the magnetic field.

Solution

The magnitude of deflecting torque (the torque which makes the object rotate) acting on the bar magnet which will tend to align the bar magnet parallel to the direction of the uniform magnetic field $\vec{B}$ is

$$\left|\vec{\tau}\right| = p_m B\sin\theta$$

The magnitude of restoring torque acting on the bar magnet can be written as

$$\left|\vec{\tau}\right| = I\frac{d^2\theta}{dt^2}$$

Under equilibrium conditions, both magnitude of deflecting torque and restoring torque will be equal but act in the opposite directions, which means

$$I\frac{d^2\theta}{dt^2} = -p_m B\sin\theta$$



(a) Why a freely suspended bar magnet in your laboratory experiences only torque (rotational motion) but not any translatory motion even though Earth has non-uniform magnetic field?

It is because Earth's magnetic field is locally(physics laboratory)uniform.

(b) Suppose we keep a freely suspended bar magnet in a non-uniform magnetic field. What will happen?

It will undergo translatory motion (net force) and rotational motion (torque).

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